Mahiraj Seervi

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May 2028 IITR Graduate with a passion for Designing Robots

Education

8.508/10 **B.Tech. in Mechanical Engineering**, Indian Institute of Technology Roorkee 2024 – 2028
92.6% Class XII (CBSE), G.S. Jangid Memorial School, Jodhpur 2024
96.0% Class X (CBSE), Mahaveer Public School, Jodhpur 2022

Relevant Coursework: CORE: Mathematics I–II, Numerical Analysis, Fluid Mechanics, Mechanics of Materials, AI/ML ; Robotics & Design: Kinematics of Mechanisms, Theory of Machines, Manufacturing Processes, Engineering Drawing

Experience

Research Intern | **Stochastic Robotics Lab, Indian Institute of Science (IISc)** **May 2026 – Present**
Actuator Co-Design & Optimization for Legged and Humanoid Robots | *Prof. Shishir N. Y. Kolathaya* *Bengaluru, India*

- Developed an **automated actuator design framework** for **Internal planetary (single, compound, dual-stage)** and **Wolfrom (WPG)** gearboxes for **frameless inrunner/outrunner BLDC motors**, maximizing **torque density** in **minimal axial width**
- Replaced **heuristic-based** selection with an **optimizer** using **Modified Lewis gear-strength model** and a mass-width-efficiency cost function to select the **optimal actuator** for any target gear ratio and motor, auto-generating CAD across **150+ parameters**
- Designing and manufacturing a **metal dual-stage inrunner planetary actuator** for high-torque, thermally robust **humanoid** joints
- Designed and fabricated a chain-driven **sandwich-leg monoped test bench** to validate actuators under **impact and dynamic jumping loads**, with low leg inertia and a modular **quick-swap coupling** that exchanges actuators without disassembling the leg

Projects

Autonomous Warehouse Robot for Rack Inventory Management | *Eternal, Inter IIT Tech Meet 14.0* **Nov 2025 – Dec 2025**

- Turned Eternal's rack-scanning problem into a lightweight, retractable **4-stage cascade lift** mechanism reaching **1.8 m** height
- Achieved **4 cm horizontal** / **1 cm vertical** scanning accuracy with a rigid chassis and omnidirectional **mecanum-wheel drive**
- Engineered a bearing-based **anti-slip timing-belt tensioner**, a reliability-vs-cost trade-off that cut long-term maintenance cost

3D-Printed Non-Pinwheel Cycloidal Actuator for Robotic Arm Joints | *Team Robocon IITR* **Apr 2025 – Dec 2025**

- Designed and fabricated a **high-torque-density** 3D-printed **non-pinwheel cycloidal drive** (**15:1**, **1.08 kg**) using a **DC motor**
- Delivered **120 N·m** torque with $\leq$**0.5°** **backlash** and full **backdrivability** in a **lightweight build** at a fraction of commercial cost
- Diagnosed eccentric-shaft failure in **load testing** and reinforced the PLA shaft with a triangular bolt pattern, **raising load capacity**

Modular Multi-Actuator Quadruped Testbed for Drivetrain Benchmarking | *Prof. P. M. Pathak, IIT Roorkee* **Feb 2026 – Present**

- Architected a **12-DOF quadruped** to benchmark **QDD planetary, cycloidal, harmonic and capstan** drives on a single platform
- Evaluating drivetrains on **torque density, efficiency, backdrivability and impact resilience** for dynamic **legged locomotion**
- Integrated **compliant leg linkages** for impact absorption and ground-contact stability, validating range of motion and gait geometry

High-Payload 6-DOF Robotic Manipulator – Gripper & Wrist | *Team Robocon IITR* **Aug 2025 – Jan 2026**

- Designed a self-centering **parallel lead-screw gripper** with an **11 cm** stroke for **precise, symmetric** grasping of heavy payloads
- Built a compact **2-DOF fully differential wrist** with 3D-printed **bevel gears**, sharing load across both motors for pitch and roll
- Integrated the gripper–wrist module with the **6-DOF arm**, reliably handling **9–10 kg payloads** with a maximum reach of **1.2 m**

6-Wheel Mars Rover with Custom Mantis Suspension | *Team Robocon IITR* **Apr 2025 – Jan 2026**

- Designed a custom **6-wheel mantis suspension** rover with deformable **TPU wheels** to improve traction on rough Martian terrain
- Developed an **auger-based soil sampler** on a **parallel-linkage mechanism** driven by **linear actuators**, achieving **15 cm** dig depth
- Built an **anti-contamination** sample-drop mechanism feeding soil into an onboard **colorimetric analysis** unit for life detection

Technical Skills

CAD & Design: SolidWorks, Onshape (API automation)
Programming: Python, C++
Simulation: ANSYS, MATLAB
Manufacturing: 3D Printing (FDM), LMC, Rapid Prototyping

Achievements

International Rank 2 **International Rover Challenge (IRC)** – IDMO arm task; **International Rank 11** overall 2026
International Top 10 **European Rover Challenge (ERC)** – remote edition, representing Team Robocon IITR 2025
1st among 350+ **SolidWorks CAD Competition** – intra-college design challenge, IIT Roorkee 2025
2nd Overall **Tech Exhibition, Srishti** – IIT Roorkee's annual technical exhibition, Team Robocon 2026

Leadership & Extracurriculars

Mechanical Lead | *Team Robocon IITR* **Apr 2026 – Present**

- Leading the **40+ member** mechanical division, owning rover architecture, subsystem design reviews and fabrication planning
- Contributed to the redesign of the rover arm for the IDMO mission, taking the team from **12th to 2nd globally** at IRC 2026

Flight Engineer | *Team India* **Sep 2026 – Nov 2026**

- Represented Team India (25+ members) as Flight Engineer at the International Space Games, a Mars simulation in Russia
- Led on-the-spot repairs of power, life-support and robotics; designed rover soil-sampling and solar-panel cleaning mechanisms